

[CASE REPORT]

Spontaneous Remission in a Patient with Anti-OJ Autoantibodies-positive Anti-synthetase Syndrome Following SARS-CoV-2 Infection and mRNA Vaccination

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Abstract:

Anti-synthetase syndrome (ASyS) is a subset of idiopathic inflammatory myopathies characterized by a triad of myositis, interstitial lung disease, and arthritis. Patients with ASyS are generally treated with glucocorticoids, immunosuppressants, or both. We encountered a 53-year-old woman who developed anti-OJ autoantibodies-positive anti-synthetase syndrome following severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection and CoV-2 messenger RNA (mRNA) vaccination. Her dyspnea and rash resolved within 4 weeks of the initial examination, achieving spontaneous remission without treatment 52 weeks after the diagnosis. This case suggests a potential association between SARS-CoV-2 infection, mRNA vaccination, and the transient development of anti-OJ antibodies-positive ASyS.

Key words: anti-synthetase syndrome, SARS-CoV-2 infection, SARS-CoV-2 mRNA vaccination, spontaneous remission

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Introduction

Anti-synthetase syndrome (ASyS) is a subset of idiopathic inflammatory myopathy (IIM), characterized by the presence of anti-aminoacyl-transfer RNA (tRNA) synthetase (anti-ARS) antibodies and a triad of myositis, interstitial lung disease (ILD), and arthritis (1-3). One of the anti-ARS antibodies, anti-OJ antibodies, which recognizes isoleucyl tRNA synthetase (4), has been associated with a high risk of developing active ILD in positive patients (5-7).

Coronavirus disease 2019 (COVID-19) is an infectious disease caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Several cases of ASyS occurring after COVID-19 episodes or SARS-CoV-2 messenger RNA (mRNA) vaccination have been documented in recent years (8-11).

We herein report a case of an anti-OJ antibodies-positive

patient who developed ASyS following SARS-CoV-2 infection and mRNA vaccination. Interestingly, the patient achieved spontaneous remission with decreased levels of anti-OJ antibodies, without any treatment.

Case Report

The patient was a 53-year-old woman with a history of mild COVID-19 5 months prior to consultation. Common cold symptoms associated with COVID-19 resolve within a few days. However, she experienced mild dyspnea on exertion four weeks after COVID-19 (four months prior to consultation). The SARS-CoV-2 test [polymerase chain reaction] at that time was negative. She received the third dose of the SARS-CoV2 mRNA vaccine [BNT162b2; Pfizer-Biopharmaceutical New Technologies (BioNTech)] as scheduled, considering that 14 weeks had passed since the last SARS-CoV-2 infection. Her dyspnea and dry cough wors-

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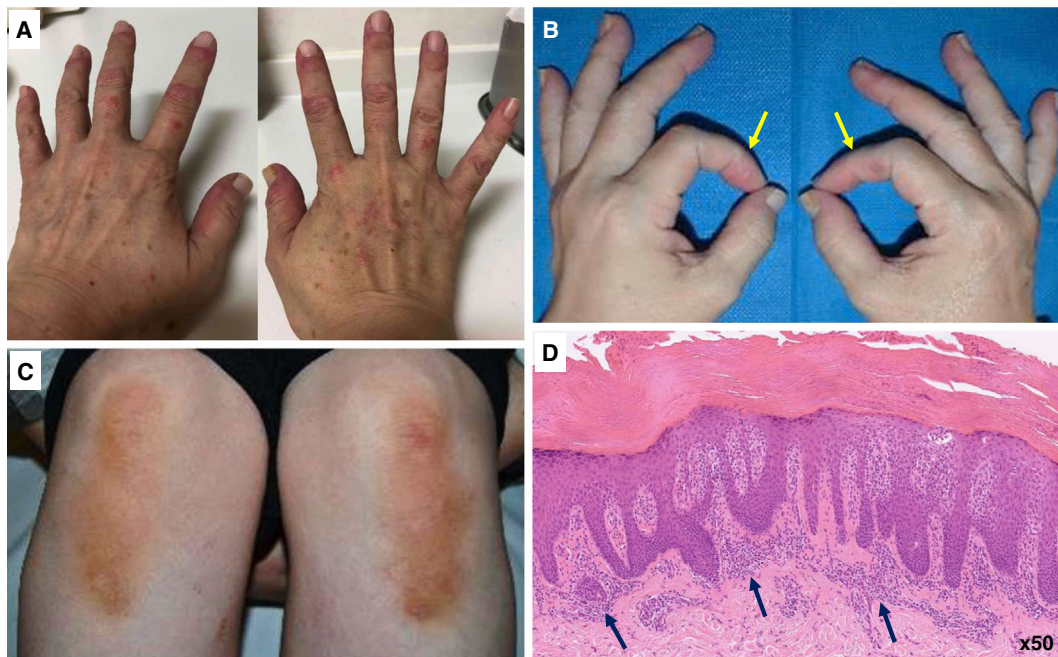


Figure 1. Dermatologic findings at the first visit. A physical examination revealed the skin manifestations of dermatomyositis. (A) Gottron's papules and periungual erythema, (B) mechanic's hands, and (C) Gottron's sign (knees). (D) A skin biopsy of the finger erythema showing hyperkeratosis, parakeratosis, liquefaction degeneration in the epidermis, and perivascular lymphocyte infiltration around small vessels in the upper dermis (arrows) (Hematoxylin and Eosin staining, $\times 50$)

ened two weeks after vaccination (four weeks before consultation).

She was referred to a medical institution, where chest computed tomography (CT) revealed diffuse consolidation in the bilateral lower lung lobes. Erythema was observed on the fingers of both hands and extensor surfaces of both elbows, and laboratory testing demonstrated elevated myogenic enzyme levels. These findings suggested dermatomyositis onset, and the patient was referred to our department for a further examination.

At her first visit to our department (week 0), vital signs revealed a body temperature of 36.6°C , respiratory rate of 16 cycles/min, and oxygen saturation (SpO_2) of 96% on room air, which decreased to 89% (Borg Dyspnea Scale 3) after a 6-minute walk test. She exhibited Gottron's papules, mechanic's hands, periungual erythema in the fingers of both hands, and Gottron's sign over both knees (Fig. 1A-C). A skin biopsy of the Gottron's papules confirmed hyperkeratosis, parakeratosis, and liquefaction degeneration, with the presence of perivascular lymphocyte infiltrates in the upper dermis (Fig. 1D), all of which were consistent with dermatomyositis. Although the patient had no muscle pain on grasping or muscular weakness, small joint arthritis was observed at the two proximal interphalangeal joints of both hands. Chest auscultation revealed fine crackles in the bilateral lower lung fields.

Laboratory tests revealed elevated levels of a myogenic enzyme [creatinine kinase (CK): 447 (reference range: 41-153) U/mL] and ILD-associated markers [Krebs von den Lungen (KL)-6:2366 (105-401) U/mL; surfactant protein

(SP)-D: 434 (<110) ng/mL]. Anti-nuclear antibody (ANA) testing using indirect immunofluorescence confirmed a cytoplasmic pattern, and immunoprecipitation revealed the presence of anti-OJ antibodies (Table 1).

Chest radiography and high-resolution (HR)-CT showed extensive consolidation and ground-glass opacities, mainly in the bilateral lower lung lobes (Fig. 2A, B). During bronchoscopy, lymphocyte-predominant bronchoalveolar lavage fluid was collected (total cell counts: $1.8 \times 10^5/\text{mL}$, lymphocytes: 54.0%, macrophages: 42.0%, neutrophils: 0.5%, eosinophils: 3.5%). Severe lymphocytic infiltration around the alveolar septa and polypoid fibrosis (Masson's bodies) were observed in the transbronchial lung biopsy tissue (Fig. 2C, D). Collectively, these findings were considered consistent with the cellular non-specific interstitial pneumonia and organizing pneumonia patterns.

Following the American College of Rheumatology/European League Against Rheumatism (ACR/EULAR) 2017 classification criteria for IIM, the patient had a total of 8.8 points (age of onset ≥ 40 years old: 2.1 points, Gottron's papules/sign: 5.4 points, elevated serum levels of CK: 1.3 points) (12). Given the presence of anti-ARS antibodies and clinical symptoms of ILD, arthritis, and mechanic's hands, she was diagnosed with ASyS, adhering to the ASyS classification criteria proposed by Solomon et al. (13).

Given the disease activity, intensive immunosuppressive therapy with high-dose glucocorticoids (GCs) was planned for the initial treatment. However, the patient noted apparent relief of dermatological and respiratory symptoms two weeks before the final diagnosis. She was placed under

Table 1. Laboratory Findings at the First Visit.

CBC		Immunology	
WBC	5,300 / μ L	ANA (titer)	<1:40
Neutrophils	65.0 %	Cytoplasmic pattern	(+)
Lymphocytes	27.0 %	Anti-ARS Abs	<5.0 U/mL
Eosinophils	4.0 %	Anti-MDA5 Abs	<4.0 U/mL
Monocytes	4.0 %	Anti-mitochondrial M2 Abs	<5.0 U/mL
RBC	4.49 $\times$ 10 ⁶ / μ L	Anti-SS-A Abs	2.8 U/mL
Hb	13.1 g/dL	Anti-SS-B Abs	2.9 U/mL
Plt	21.7 $\times$ 10 ⁴ / μ L	Anti-Ro52 Abs	0.15 U/mL
Biochemistry		Immunoprecipitation	
TP/Alb	5.9/3.6 g/dL	Anti-OJ Abs	(+)
AST/ALT	20/20 U/L		
LDH	195 U/L	Infection	
BUN/Cre	10/0.6 mg/dL	SARS-CoV-2 RT-PCR	(-)
Na/K/Cl	139/4.2/108 mmol/L	β -D-glucan	<6.0 pg/mL
CK	447 U/L	QuantiFERON TB gold plus	(-)
Aldolase	6.9 U/L		
KL-6	2,366 U/mL	Arterial blood gas (room air)	
SP-D	434 ng/mL	pH	7.40
Ferritin	30 ng/mL	PCO ₂	36.6 mmHg
CRP	0.1 mg/dL	PO ₂	91 mmHg
ESR	12 mm/h	Base excess	0.2 mmol/L
		HCO ₃ ⁻	24.0 mmol/L

CBC: complete blood count, WBC: white blood cells, RBC: red blood cells, Hb: hemoglobin, Plt: platelets, TP: total protein, Alb: albumin, AST: aspartate aminotransferase, ALT: alanine aminotransferase, LDH: lactate dehydrogenase, BUN: blood urea nitrogen, Cre: creatinine, CK: creatine kinase, KL-6: krebs von den lungen -6, SP-D: surfactant protein-D, CRP: C-reactive protein, ESR: erythrocyte sedimentation rate, ARS: aminoacyl t-RNA synthetases, Abs: antibodies, MDA5: melanoma differentiation-associated gene 5, RT-PCR: reverse transcription polymerase chain reaction, TB: tuberculosis

follow-up observation without treatment, and her symptoms improved significantly over the following weeks. At 52 weeks after the diagnosis, a follow-up examination demonstrated normalized levels of CK, KL-6, and SP-D and resolution of interstitial lung shadows on HR-CT (Fig. 3). Furthermore, the ANA test performed on week 52 was negative for the cytoplasmic pattern, and anti-OJ antibodies levels decreased to a fifth of those observed at the time of the diagnosis (Fig. 4). The patient has remained in remission for over a year without any relapse of symptoms or increased autoantibodies levels.

Discussion

Environmental factors (e.g. viral or bacterial infections) (14) and malignant tumors (15) have been implicated as pathogenic triggers of ASyS. It is hypothesized that these triggers lead to the extracellular release of ARS antigens, which induces excessive immune responses (innate and acquired immunity), resulting in anti-ARS Ab production and organ dysfunction (16-20). Although ASyS is generally treated with GCs and immunosuppressants, there has been only one reported case of spontaneous remission (21).

Several studies have linked ASyS to SARS-CoV-2 infection (9, 11, 22-27) and SARS-CoV-2 mRNA vaccina-

tion (28, 29). According to these reports, the pathogenic mechanisms of immune responses and cytokine production are induced by these triggers, leading to anti-ARS antibodies production and organ dysfunction (29, 30). Table 2 summarizes the clinical characteristics of ASyS cases following SARS-CoV-2 infection, SARS-CoV-2 mRNA vaccination, or both infection and vaccination. To our knowledge, this is the first reported case of ASyS triggered by an infection and exacerbated by vaccination. However, given the limited number of reports and the use of high-dose GC therapy in these patients, the specific impact of COVID-19 episodes and SARS-CoV-2 mRNA vaccination on ASyS pathogenesis remains unclear.

In the present case, the patient had typical features of ASyS, including a skin rash (Gottron's papules and signs), elevated myogenic enzyme levels, ILD, and positive anti-OJ antibodies. These findings fulfilled the ACR/EULAR 2017 classification criteria for IIM and the ASyS classification criteria. In addition, transbronchial lung and skin biopsy findings were consistent with those in previous reports of anti-ARS (OJ) antibodies-positive patients (5, 31). As such, the findings in this patient are consistent with new-onset ASyS and cannot be explained by Long COVID.

Our case differed from typical ASyS cases in two respects. First, the patient had a history of SARS-CoV-2 infec-

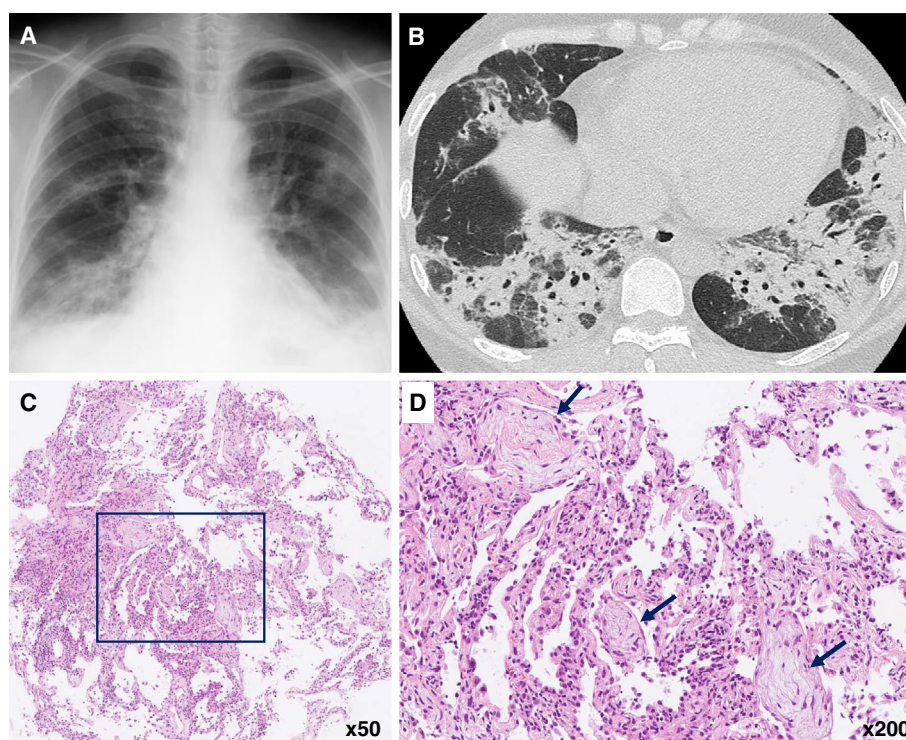


Figure 2. Chest X-ray, high-resolution CT, and TBLB findings at the first visit. (A) Chest X-ray and (B) chest high-resolution CT. (C) Tissue from the TBLB showed lymphocyte infiltrate and mild fibrosis in the alveolar septa [Hematoxylin and Eosin (H&E) staining, $\times 50$]. (D) Tissue from the TBLB showed polypoid plugs of loose organizing connective tissue (Masson bodies, arrows) within the alveolar tissue (H&E staining, $\times 200$). CT: computed tomography, TBLB: transbronchial lung biopsy

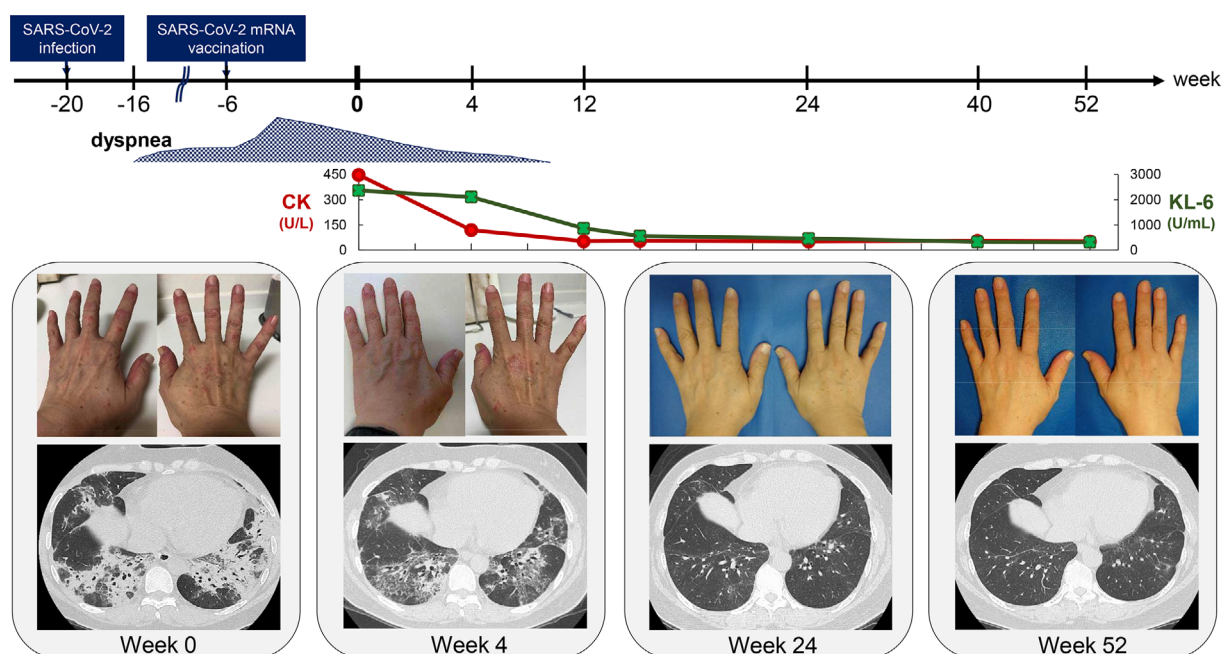


Figure 3. Clinical course. The skin rashes and dyspnea, as well as CK, KL-6, and interstitial lung abnormalities on CT seen at the first visit (week 0), were in complete remission by week 52 without any treatment. CK: creatine kinase, KL-6: krebs von den lungen-6

tion and SARS-CoV-2 mRNA vaccination before the onset of ASyS. ILD-associated dyspnea was reported four weeks following infection, exhibiting a clinical course similar to that of the aforementioned cases (Table 2). Furthermore, the

development of a skin rash, elevated CK levels, and worsening dyspnea were observed immediately after mRNA vaccination. Second, the patient achieved complete resolution of all organ dysfunctions and a substantial reduction in anti-OJ

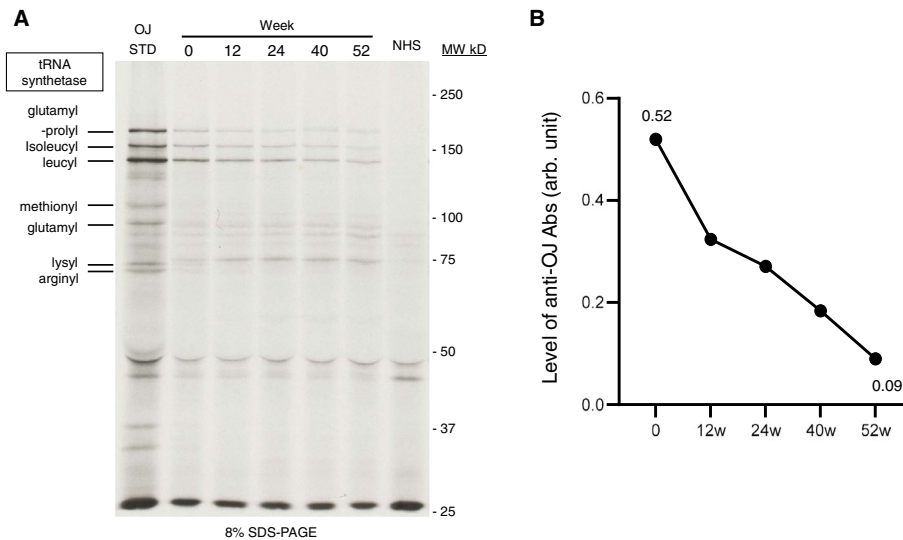


Figure 4. Detection and time course of the level of anti-OJ antibodies by immunoprecipitation. (A) 35 S-methionine/cysteine-labelled K562 cell extract was immunoprecipitated by sequential sera from the patient (Lanes 2–5). Lane 1: standard (STD), anti-OJ reference serum, Lane 2: at the first visit (week 0), Lane 3: week 12, Lane 4: week 24, Lane 5: week 52, Lane 6: normal human serum (NHS). (B) Time course of anti-OJ antibodies levels. The bands of leucyl tRNA synthetase using anti-OJ standard (STD) serum and sequential sera from the present case were quantified using the Image J software program. The volume of the leucyl tRNA synthetase band of STD was defined as 1 arbitrary unit (32). SDS-PAGE: sodium dodecyl sulfate-polyacrylamide gel electrophoresis, Abs: antibodies, arb: arbitrary

Table 2. Clinical Characteristics of the Patients with ASyS after SARS-CoV-2 Infection or MRNA Vaccination.

	After SARS-CoV-2 infection (9,11,22-27)	After SARS-CoV-2 mRNA vaccination (28,29)	After SARS-CoV-2 infection and vaccination
Number of cases	8	3	1 (Our case)
Age, mean $\pm$ SD, years	51.5 $\pm$ 14.6	44.7 $\pm$ 13.9	53
Female, n (%)	1 (12.5)	3 (100)	1 (100)
Duration from onset, mean $\pm$ SD, days	29.8 $\pm$ 13.2	$\approx$ 10	28 (from infection) /14 (from vaccination)
Autoantibodies, (n)	Jo-1 (2), EJ (2), OJ (1), PL-7 (2), Jo-1+PL-12 (1)	Jo-1 (1), PL-7 (1), PL-12 (1)	OJ
Organ manifestation, (n)	ILD (8), myositis (6), arthritis (5), rash (3)	ILD (1), myositis (1), arthritis (1), unknown (2)	ILD, myositis, rash
Treatment, (n)	High-dose GC (8), IVCY (3), other ISD (7)	High-dose GC (2), ISD (2), unknown (1)	None
Outcome, (n)	Effective (8)	Effective (1), poor (1: relapsed), unknown (1)	Spontaneous remission

ASyS: anti-synthetase syndrome, SARS-CoV-2: severe acute respiratory syndrome coronavirus 2, mRNA: messenger RNA, SD: standard deviation, ILD: interstitial lung disease, GC: glucocorticoid, ISD: immunosuppressive drugs

Ab levels without treatment.

In conclusion, our findings strongly suggest that, in the present case, ASyS was initially triggered by SARS-CoV-2 infection and exacerbated by SARS-CoV-2 mRNA vaccination, all of which contributed to the transient development of ASyS with anti-OJ antibodies.

Author's disclosure of potential Conflicts of Interest (COI).

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